

SDG Goal 3

Good health and well-being

SDG Target 3.9

By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination

SDG Indicator 3.9.2

Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe Water, Sanitation and Hygiene for All (WASH) services)

Time series

Mortality attributed to contact to unsafe water, unsafe sanitation and lack of hygiene

1. General information on the time series

- Date of national metadata: April 30, 2026
- National data: <https://sdg-indicators.de/3-9-2>
- Definition: The time series shows the number of deaths attributed to contact to unsafe water, unsafe sanitation and lack of hygiene per 100,000 inhabitants.

Classification is based on ICD-10 codes. ICD-10 is the 10th Revision of the International Statistical Classification of Diseases and Related Health Problems. In its German Modification (ICD-10-GM), it serves as the official classification system for coding diagnoses in outpatient and inpatient care in Germany. The following ICD-10 codes are included:

(a) Acute respiratory infections:

H65: Nonsuppurative otitis media
H66: Suppurative and unspecified otitis media
J00-J06: Acute upper respiratory infections
J09-J18: Influenza and pneumonia
J20-J22: Other acute lower respiratory infections
P23: Congenital pneumonia
U04: Severe acute respiratory syndrome (SARS)

(b) Fractions of diarrhoea:

A00-A09: Intestinal infectious diseases

(excluding: **A02:** Other salmonella infections and **A05:** Other bacterial food-borne intoxications)

(c) Intestinal nematode infections:

B76: Hookworm diseases
B77: Ascariasis
B79: Trichuriasis

(d) Protein-energy malnutrition:

E40-E46: Malnutrition

- Disaggregation: age group, sex, type disease

2. Comparability with the UN metadata

- Date of UN metadata: July 2022
- UN metadata: <https://unstats.un.org/sdgs/metadata/files/Metadata-03-09-02.pdf>
- The time series is compliant with the UN metadata.

3. Data description

- The number of deaths is taken from the Federal Statistical Office's statistics on causes of death and is based on an analysis of death certificates issued by medical doctors.

For the calculation of population data, the retroactively calculated average population provided by the Federal Statistical Office is used. For 2010, this is based on the 2011 census as well as migration, birth, and death statistics; for the years 2012 to 2021, the calculation is based on the 2022 census.

4. Access to data source

- GBE – Table: Deaths, Mortality figures [Selections: Hierarchy: ICD-10 A00-T98 (3-Steller), ICD-10 U00-U99 (3-Steller); ICD10: A00, A01, A03, A04, A06-A09, B76-B77, B79, E40-E46, H65-H66, J00-J22, P23, U04]:
https://www.gbe-bund.de/gbe/isgbe.fundstellen?p_uid=gast&p_aid=99629820&p_sprache=E&p_thema_id=3900&p_action=TRT
- Average population – GENESIS online 12411-0041:
<https://www-genesis.destatis.de/genesis//online?operation=table&code=12411-0041&bypass=true&levelindex=1&levelid=1639396599054#abreadcrumb>
- Population data based on Census 2022 (only available in German):
https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Bevoelkerung/Bevoelkerungsstand/_inhalt.html#sprg233540

5. Metadata on source data

- Quality Report – Causes of death statistics (only available in German):
<https://www.destatis.de/DE/Methoden/Qualitaet/Qualitaetsberichte/Gesundheit/todesursachen.pdf>
- Quality Report – Microcensus (only available in German):
<https://www.destatis.de/DE/Methoden/Qualitaet/Qualitaetsberichte/Bevoelkerung/einfuehrung.html>

6. Timeliness and frequency

- Timeliness: t + 8.5 months
- Frequency: Annual

7. Calculation method

- Unit of measurement: Per 100,000 inhabitants
- Calculation:

$$\text{Mortality rate}_i = \frac{\sum_i \text{Deaths with assigned ICDk}_i \text{ [number]}}{\text{Population [number]}} \cdot 100,000$$

With: $i \in \{\text{Acute respiratory infections; Fractions of diarrhoea; Intestinal nematode infections; Protein-energy malnutrition}\}$

$k_{\text{Acute respiratory infections}} \in \{H65; H66; J00; \dots; J22; P23; U04\}$

$k_{\text{Fractions of diarrhoea}} \in \{A00; A01; A03; A04; A06; \dots; A09\}$

$k_{\text{Intestinal nematode infections}} \in \{B76; B77; B79\}$

$k_{\text{Protein-energy malnutrition}} \in \{E40; \dots; E46\}$